SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based
COURSE NAME: Cloud Computing and Big Data Analytics
COURSE CODE: CSA15
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)
Dave's Taxonomy: Manipulation (Level 2) Question Count: 5 Total Marks: 25

Q1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads. [5 Marks]

Answer:

Cloud Architecture Design

The streaming platform should be built on a horizontally scalable microservices architecture deployed across multiple availability zones. Video content is stored in object storage (e.g., Amazon S3) and distributed globally through a Content Delivery Network (CDN) with edge caching, so popular videos are served from cache nodes close to users rather than the origin server, drastically cutting buffering and latency. An origin-shield layer and multi-tier caching (edge cache, regional cache, origin cache) reduce repeated load on backend servers.

Load Balancing and Auto-Scaling

An elastic load balancer distributes incoming user requests (e.g., stream requests, API calls) evenly across a pool of application servers, preventing any single server from becoming a bottleneck. Auto-scaling groups monitor real-time metrics such as CPU utilization and concurrent connections, automatically adding server instances during peak viewing hours (e.g., a new show release) and scaling down during off-peak times, ensuring smooth, uninterrupted streaming without over-provisioning.

Cost Optimization Strategies

- Use tiered storage: keep frequently watched content in higher-cost, high-performance storage and archive rarely watched content to cheaper cold storage.
- Leverage reserved or spot instances for predictable and batch transcoding workloads to reduce compute costs.
- Maximize CDN cache-hit ratio to minimize expensive origin-server bandwidth usage.
- Use adaptive bitrate streaming to reduce bandwidth costs while maintaining acceptable quality based on each user's network conditions.

Q2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model. [5 Marks]

Answer:

Proposed Solution: Container Orchestration and Multi-Cloud Strategy

The startup should standardize all deployments using Kubernetes, which can run consistently across AWS, Azure, and GCP, ensuring the same container images, configuration manifests (YAML), and deployment pipelines work identically regardless of provider. A GitOps-based CI/CD pipeline (e.g., using ArgoCD or Flux) can automate consistent deployment across clusters in different clouds, reducing configuration drift and human error.

Service Discovery and Scaling

Kubernetes' built-in service discovery automatically assigns each microservice a stable internal DNS name and IP, so services can locate and communicate with each other even as individual pod instances are created or destroyed. Horizontal Pod Autoscaling (HPA) monitors CPU/memory or custom metrics and automatically increases or decreases the number of running container replicas to match load, ensuring the fintech application scales efficiently during transaction spikes across every cloud cluster.

Benefits and Risks of Multi-Cloud Deployment

- **Benefits:** Avoids vendor lock-in, improves resilience (if one provider has an outage, traffic shifts to another), and allows the startup to choose the best-priced or best-performing services from each provider.
- **Risks:** Increased operational complexity from managing multiple environments, potential data consistency and latency issues between clouds, higher networking costs for cross-cloud traffic, and the need for specialized skills to manage several providers' distinct services and security models.

Q3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures. [5 Marks]

Answer:

Backup, Replication, and Versioning Design

The platform should enable automated, scheduled backups of databases and file storage using cloud-native backup services (e.g., AWS Backup, Azure Backup), storing snapshots in a separate, isolated storage account to protect against accidental deletion. Object versioning should be enabled on cloud storage buckets, so every previous version of a file is retained and can be restored even if the current version is deleted or overwritten. Cross-region replication ensures a secondary copy of data exists in a geographically separate location.

Achieving RTO and RPO

Recovery Point Objective (RPO), the maximum acceptable amount of data loss measured in time, is achieved by taking frequent incremental backups and enabling continuous or near-real-time replication, so that in the event of a failure, only a small window of recent data (e.g., minutes) is at risk. Recovery Time Objective (RTO), the maximum acceptable downtime, is achieved by maintaining warm standby infrastructure or automated recovery scripts/runbooks that can restore service quickly, combined with regularly tested failover procedures to minimize manual recovery steps.

Steps to Ensure Business Continuity

- Implement automated, versioned, and geographically redundant backups on a defined schedule.
- Enable soft-delete and object versioning so accidental deletions can be reversed within a retention window.

- Maintain a documented disaster recovery (DR) plan with clearly defined RTO/RPO targets for each system.
- Regularly test backup restoration and failover procedures through scheduled DR drills.
- Set up monitoring and alerts for backup job failures so issues are caught before they affect recovery capability.

Q4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments. [5 Marks]

Answer:

Hybrid Architecture Design

Sensitive workloads and regulated data remain on-premises in the agency's private datacenter, while non-sensitive, scalable workloads (e.g., public-facing citizen portals) are deployed in the public cloud. A secure, dedicated connection such as a VPN tunnel or a private leased line (e.g., AWS Direct Connect, Azure ExpressRoute) links the on-premises datacenter to the cloud environment, avoiding exposure over the public internet, with a hybrid gateway managing traffic routing between the two environments.

Identity Management and Encryption

A federated identity management system (e.g., Active Directory Federation Services integrated with Azure AD) should be implemented so users authenticate once and are consistently recognized across both on-premises and cloud systems, with role-based access control enforcing least-privilege access to sensitive data. All data must be encrypted at rest using strong encryption standards (e.g., AES-256) and in transit using TLS across the VPN/dedicated link, with encryption keys managed centrally through a key management service, ensuring sensitive government data is protected throughout its lifecycle.

Challenges in Managing Hybrid Cloud Environments

- Maintaining consistent security policies and access controls across two fundamentally different environments (on-premises and cloud).
- Network complexity and potential latency in communication between on-premises and cloud components.
- Ensuring compliance and data sovereignty when data or metadata could inadvertently cross environment boundaries.
- Increased operational overhead from managing, patching, and monitoring two separate infrastructure stacks with different toolsets.

Q5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection. [5 Marks]

Answer:

Multi-Region Deployment Strategy

The SaaS application should be deployed as an active-active or active-passive architecture across at least two geographically separate cloud regions, with application servers, load balancers, and databases replicated in each region. A global traffic manager or DNS-based routing service (e.g., AWS Route 53, Azure Traffic Manager) directs users to the nearest healthy region under normal conditions and automatically reroutes traffic away from a failed region.

Data Replication and Failover Configuration

Databases should use synchronous or near-real-time asynchronous replication between regions (e.g., a primary database in Region A continuously replicating to a standby in Region B), ensuring the standby has up-to-date data ready to take over. Automated health checks continuously probe each region's availability; if the primary region fails, a failover mechanism promotes the standby region to primary and updates DNS/traffic routing within seconds to minutes, minimizing service disruption without manual intervention.

Monitoring and Alerting Strategies

- Deploy centralized monitoring (e.g., CloudWatch, Azure Monitor, Prometheus/Grafana) tracking application health, latency, and error rates across all regions.
- Configure automated alerts for threshold breaches (e.g., high error rate, region unreachable) sent to on-call engineering teams via SMS/email/Slack.
- Implement synthetic transaction monitoring that periodically simulates real user actions to proactively detect degraded performance before customers are affected.
- Maintain dashboards showing real-time replication lag and failover readiness so teams can catch replication issues before they impact recovery capability.

Total Marks Awarded: 25 / 25